AI Voice App — Optimization Implementation Report

InterviewCopilot · controlled architecture optimization · 9 files changed, +584 / −163

Baseline	`7a375cf "foundation" — verified byte-for-byte against the audit before any edit`
Date	23 August 2026
Scope	Targeted optimization of the boundaries the audit identified. **No rewrite.** The STT dispatcher/consumer, question-gate architecture, barge-in, follow-up deferral, cancellation and error-recovery paths are untouched.
Verification	Regression suite run after *every* change: 33-stream VAD corpus · 1,562-transcript gate+filter fingerprint · 6-scenario end-to-end · 26-scenario behaviour suite · 5-mode fault injection · 350-interaction resource probe · judged answer-quality benchmark · live full-stack test against a real uvicorn server with real Groq calls

Headline results — all measured, none estimated

100% VAD speech-detection recall (was 95.7%)	**20%** VAD false-trigger rate (was 50%)	**4.1** questions/min before throttle (was 2.9)	**1.18s** end-of-speech → answer (was 1.32s)	**96.4%** judged answer quality (was 92.9% / 91.1%)

IMPLEMENTED shipped and regression-tested · **NOT IMPLEMENTED** deliberately skipped, with reason · **REJECTED** tested and disproven · **FUTURE** deferred, with the blocker named

1 · Original vs New Architecture

The pipeline shape is unchanged. Five boundaries inside it changed, and one new observability path was added. Everything the audit scored 8–9/10 was left alone.

```
                  ORIGINAL                                     NEW
─────────────────────────────────────────   ─────────────────────────────────────────
loopback capture → 20ms frames              unchanged
      │                                             │
      ▼                                             ▼
SpeechSegmenter                             SpeechSegmenter
  bool speech_active                          VadState: CALIBRATING / SILENCE /
  trigger = max(500, noise×2.5)                 NOISE / SPEECH_STARTED /
  floor 500 defeats adaptation                  SPEECH_CONTINUING / SPEECH_ENDED
  latches on noisy lines, never               trigger = max(150, noise×2.5)
  re-measures (30s escape hatch)              1s calibration BEFORE any decision
  no telemetry when it rejects                sub_threshold_runs counter + WARNING
      │                                             │
      ▼                                             ▼
dispatcher / consumer, Semaphore(3)         UNCHANGED — audit rated this the
                                            strongest code in the repo
      │                                             │
      ▼                                             ▼
transcript_quality                          transcript_quality
  _WORD_RE = [a-z][a-z']*                     Unicode-aware; vowel heuristic
  non-Latin → ratio 0 → DROPPED               confined to Latin tokens only
      │                                             │
      ▼                                             ▼
QuestionAnswerPipeline                      QuestionAnswerPipeline
  sleep(180ms) THEN evaluate                  settled? sleep(30ms) : sleep(180ms)
  timer can overrule active speech            speech_active outranks the timer
  _classify_question re-run each cycle        memoised per exact transcript
  get_context THEN get_history                asyncio.gather
  empty answer → answer_done (success)        empty answer → error + WARNING
      │                                             │
      ▼                                             ▼
system prompt 1,978 tok/question            system prompt 1,186 tok/question
      │                                             │
      ▼                                             ▼
answer_hub: mkdir+open+write+close          answer_hub: cached line-buffered
per event, on the event loop                handle; same durability
      │                                             │
      ▼                                             ▼
overlay: progressive token reveal            unchanged
```

Files changed, and why each one had to

File	±	Functions touched	Reason
`app/services/vad.py`	+143	`VadState` (new), `EnergySpeechDetector.__init__/is_speech`, `SpeechSegmenter.__init__`, `_observe_state` (new), `state` / `sub_threshold_runs` / `trigger_level` properties, `_build_detector`	F-02: both speech-detection defects live here
`app/services/question_gate.py`	+395/-163	`_should_keep_waiting`, `_process_after_debounce`, `_is_settled` (new), `_PendingTranscript`, `_generate_answer`, `_generate_screen_answer`, `_build_answer_messages`	F-01, F-03, F-05, F-06, F-08 + parallel store reads
`app/core/config.py`	+49	`vad_energy_threshold`, `vad_calibration_ms` (new), `question_settled_debounce_ms` (new)	Every tuned value carries its measurement, per project convention
`app/services/answer_hub.py`	+57	`__init__`, `_log_event`, `_log_handle` / `_drop_log_handle` / `close_logs` (new)	F-09: synchronous disk I/O on the event loop
`app/routes/audio_ws.py`	+18	`audio_websocket`	Wire calibration; emit the new sub-threshold telemetry
`app/services/stt/registry.py`	+27	`_build_fallback`, `_warn_fallback_unavailable` (new)	F-07: configuration implied redundancy that did not exist
`app/services/transcript_quality.py`	+25	`_WORD_RE`, `_LATIN_TOKEN_RE` (new), `lexical_word_ratio`	F-04: non-Latin scripts silently discarded
`client/test_question_gate_scenarios.py`	+25	new scenario	Permanent regression for the split-question bug
`.env.example`	+8	—	Document the two new settings

Deliberately NOT changed

`_transcription_dispatcher` / `_result_consumer` · barge-in cancellation and ownership re-check · follow-up deferral · `close_utterance` closure protocol · error-recovery paths · clause splitting and fuzzy intent matching · vendor abstraction · `llm.py` · `redis_client.py` · `overlay_app.py` · `main.py` · all STT provider adapters.

2 · Before / After — the required benchmark table

Metric	Before	After	Change	How measured
Speech detection (recall)	95.7% (22/23)	100.0% (23/23)	+4.3 pp	33-stream scored corpus
False VAD triggers	50.0% (5/10)	20.0% (2/10)	−60% rel.	same corpus, noise-only streams
Question splitting	2 answers	1 answer	fixed	log-replayed scenario
STT latency	304 ms p50	304 ms p50	unchanged	not touched by design
STT accuracy (English)	1,517 kept	1,517 kept	unchanged	1,562-transcript fingerprint
STT accuracy (non-Latin)	dropped	kept	fixed	real Devanagari transcript from logs
LLM prompt tokens/question	2,478	1,814	−27%	Groq `usage.prompt_tokens`
LLM total tokens/question	2,719	1,970	−28%	Groq `usage.total_tokens`
Questions/min before throttle	2.9	4.1	+41%	8,000 TPM ÷ measured usage
LLM TTFT (healthy)	543 ms	648 ms	≈ unchanged	live server, single sample
First response latency (e2e)	1.30–1.33 s	1.16–1.21 s	−140 ms	6-scenario offline e2e
Answer quality (judged)	92.9% / 91.1%	96.4%	within noise ↑	8 questions × 7 rules, independent judge
Error rate (failures reported)	4 of 5	5 of 5	+1	fault injection
Log write per event	553 µs	29.3 µs	−95%	400-event microbenchmark
Gate CPU on a held buffer	30.3 ms × 22	30.3 ms × 1	−636 ms	direct profiling, 156-word buffer
Barge-in / interruption	1 cancel	1 cancel	unchanged	26-scenario suite
Resource growth / 100 turns	none	none	unchanged	350-interaction probe

Two token measurements, both reported

Token counts depend on résumé/JD size, so both contexts are given rather than the flattering one. Against the audit's synthetic probe context: **2,719** → **1,970** total (2.9 → 4.1 q/min). Against the realistic résumé + JD used in the quality benchmark: **2,178** → **1,508** total (3.7 → 5.3 q/min). Both are a 28–31% reduction and a 41–43% throughput gain.

3 · Changes Implemented

`F-01` System prompt compressed 1,978 → 1,186 tokens

P0 **IMPLEMENTED**

File / function	`question_gate.py` → `_build_answer_messages`
Change	Removed three field-specific worked examples, trimmed the 14-pair plain-word substitution list to its 6 most-violated entries, merged overlapping anti-fabrication rules. Every behavioural rule kept verbatim.
Reason	The prompt was 80% of input tokens against a measured 8,000 TPM ceiling, capping the app at 2.9 questions/minute — below a real interview's rate. Past the limit answers stalled 8–17.5 s rather than failing visibly.
Benefit	**Measured: 2,719 → 1,970 tokens/question; 2.9 → 4.1 questions/minute; system prompt share of input 80% → 65%.**
Risk taken	Medium — prompt edits can silently degrade format, depth or grounding.
Tests	Judged benchmark, 8 questions × 7 rules, independent vendor (Anthropic) as judge, plus a repeat baseline run to quantify harness noise.
Result	**96.4% (54/56) vs baseline 92.9% / 91.1%. format, no_markup, no_preamble, spoken_voice, depth, answers_all all 100%. Verified live end-to-end against a real server.**

`F-02` VAD state model + calibration window + threshold floor

P1 **IMPLEMENTED**

File / function	`vad.py` → `VadState`, `EnergySpeechDetector`, `SpeechSegmenter`, `_observe_state`; `config.py`; `audio_ws.py`
Change	(a) Six-state model replacing the single `speech_active` bool. (b) A 1 s calibration window during which the detector measures the line but reports no speech. (c) Floor 500 → 150. (d) A `sub_threshold_runs` counter that logs a WARNING when audio stood above the noise floor but never cleared the trigger.
Reason	Two independent defects. The floor pinned the trigger at 500 on quiet lines, so speech below −36 dBFS produced *nothing* — no segment, no STT call, no log line. Separately, because the noise floor is estimated only from non-speech frames, a detector that latched never re-measured, and 5 of 10 noise-only streams became transcription requests.
Why both together	**This is the key design point.** Lowering the threshold alone was measured to make false triggers *worse* (50% → 60–70%). The calibration window is what makes the lower floor safe.
Benefit	**Recall 95.7% → 100%; false triggers 50% → 20%. Six states observed in one stream: `calibrating → silence → noise → speech_started → speech_continuing → speech_ended → silence`.**
Risk taken	Medium — affects every question. Mitigated: 1 s was the measured knee (2 s began missing speech already in progress at connect).
Tests	33-stream corpus; 6-scenario e2e; gate fingerprint; live server test.
Result	**All pass. Gate fingerprint unchanged. e2e latency unchanged. Calibration correctly absorbed a leading noise blip in one scenario (dropped segments 2 → 1).**

`F-03` A question is never finalized while speech is active

P1 IMPLEMENTED

File / function	`question_gate.py` → `_should_keep_waiting`
Change	Reordered the branches so the barge-in path (finished-looking question, or bare follow-up) is decided *first*, then made `speech_active` unconditionally hold any incomplete or not-obviously-finished buffer.
Reason	The soft-wait timer is a guess; the VAD is a fact. The guess could expire mid-sentence, so one spoken sentence became two questions and two answers.
Benefit	**The log-replayed case now produces exactly one answer containing "projects". Also halves token use on affected turns, compounding with F-01.**
Risk taken	Low–Medium — must not delay genuine barge-in. Branch ordering was chosen specifically to guarantee that; `question_max_wait_seconds` (8 s) still caps the hold, so a stuck VAD cannot wedge a question.
Tests	9 hand-built wait-policy cases; new permanent scenario; full 26-scenario suite; a before/after comparison run with `question_gate.py` temporarily reverted via `git stash`.
Result	**9/9 policy cases correct. Suite produced identical 28 answers and exactly 1 cancellation (barge-in) before and after. The stash comparison confirmed the four multi-answer scenarios were unaffected (8 answers both ways).**

`F-05` State-aware debounce

P1 IMPLEMENTED

File / function	`question_gate.py` → `_process_after_debounce`, `_is_settled` (new); `config.py` → `question_settled_debounce_ms`
Change	The debounce was not deleted. It became conditional: the first evaluation of a buffer the VAD has already settled (utterance closed *and* nobody talking) waits 30 ms instead of 180 ms. Every later cycle uses the full window.
Reason	The loop slept *before* evaluating, so the full debounce was paid even on questions that were already unambiguous — visible in the logs as `since_last_fragment_ms` p50 185 ms / p90 193 ms across 241 real answers.
Benefit	**e2e 1.30–1.33 s → 1.16–1.21 s. Also improves barge-in responsiveness, which was debounce-bound.**
Risk taken	Low — only short-circuits when there is nothing left to coalesce with.
Tests	e2e; full scenario suite with attention to multi-part coalescing.
Result	**All pass. `multi_part_question_in_pieces`, `midquestion_pause_merges` and all four scenario-question tests still merge correctly.**

`F-04` Unicode-aware transcript quality

P1 IMPLEMENTED

File / function	`transcript_quality.py` → `_WORD_RE`, `_LATIN_TOKEN_RE` (new), `lexical_word_ratio`
Change	Tokenise letters in any script; confine the ASCII-vowel heuristic to Latin tokens only. Non-Latin tokens count as real words and are judged by the other two signals (voiced duration, decoder confidence).
Reason	Devanagari, Tamil and CJK produced zero tokens → ratio 0.0 → dropped as `no_real_words` before ever reaching the gate.
Benefit	**Non-Latin transcripts now reach the gate. Verified on a *real Devanagari question from the production logs*: विशारो कड़ारेक माइक मनें? — previously dropped, now kept.**
Risk taken	**Medium — this filter is load-bearing for hallucination rejection.**
Tests	1,562-transcript fingerprint over the real corpus; explicit hallucination and apostrophe suites.
Result	**1,561 of 1,562 transcripts behave *identically*. The single change is the Devanagari question above. All hallucinations still rejected (`Ct.js.`, `Tch.`, `Ct-4, P. Cora?`, `.`, looping repetition). English apostrophe tokenisation bit-identical.**

A bug I introduced, caught by the regression gate, and fixed

My first F-04 attempt wrote the token pattern as `[^\W\d_][^\W\d_'’]*`. Inside a *negated* class the apostrophes are excluded rather than included, so `"I'll"` tokenised as `I` + `ll` and scored 0.33 — three English transcripts flipped to `mostly_non_words`. The fingerprint caught it immediately (`kept_by_filter` 1517 → 1516). Corrected to `[^\W\d_](?:[^\W\d_]|['’])*`, after which only the intended Devanagari transcript changed. **This is exactly why the fingerprint gate exists, and why the diff is reviewed rather than just the count.**

`F-06` Empty LLM response is a failure, not a success

P2 IMPLEMENTED

File / function	`question_gate.py` → `_generate_answer`, `_generate_screen_answer`
Change	A stream that finishes with no visible content now emits an `error` event carrying `reason: "empty_answer"` and an actionable message, plus a WARNING log naming the question. `answer_done` and `add_history` are skipped.
Reason	An empty `answer_done` showed a blank overlay with no explanation and was indistinguishable from success in the durable log. Documented in `config.py` as a real outcome on this model when reasoning consumes the whole budget.
State model	SUCCESS → `answer_done` · EMPTY → `error` + `empty_answer` · ERROR/TIMEOUT → `error` with the exception type named · CANCELLED → `answer_cancelled`. No retry loop was added.
Risk taken	Very low. The overlay's existing `error` handler already resets state to `listening` and renders the message, so no UI state is stranded.
Result	**Fault injection: 5 of 5 failure modes now report to the user (was 4 of 5), with no stuck tasks and clean recovery.**

`F-07` STT fallback reports its own absence

P2 IMPLEMENTED

File / function	`stt/registry.py` → `_build_fallback`, `_warn_fallback_unavailable` (new)
Change	When `STT_FALLBACK_ENABLED=true` but no `OPENAI_API_KEY` is configured, log one actionable WARNING. No racing, no second provider call, no new cost. Warned once per process, not per websocket.
Reason	The configuration read as though redundancy existed; it did not, and a failed transcription lost that segment silently.
Result	**Verified in the real server log during the live test. Confirmed it does not repeat on reconnect.**

`F-08 / F-09 / store reads` Three contained performance fixes

P2 IMPLEMENTED

F-08	`question_gate.py`: memoise the pure `_classify_question` per exact transcript in `gate_cache`, cleared alongside `intent_cache`. Measured: a 156-word buffer held 4 s avoided **636 ms** of CPU held under the pipeline-wide lock. Fingerprint unchanged, as a pure function must be.
F-09	`answer_hub.py`: cache a line-buffered append handle per log path instead of `mkdir` + `open` + `write` + `close` per event. **553 µs → 29.3 µs**. Durability explicitly re-verified (401/401 lines on disk after close); handle is dropped and reopened on any write failure.
Store reads	`question_gate.py`: `get_context` and `get_history` now run under `asyncio.gather`. No measurable effect on the in-memory backend; collapses two Redis round trips into one when `SESSION_STORE_BACKEND=redis`.

`NEW` **Observability for the previously invisible failure** **P1** **IMPLEMENTED**

File	`vad.py` (counter) → `audio_ws.py` (report at session close)
Change	Count sustained runs (~500 ms) of audio above the measured noise floor but below the speech trigger, and log a WARNING at session close naming the trigger level and the loudest peak reached.
Reason	The audit's hardest finding was that VAD rejection leaves *no trace at all* — no segment, no STT call, no event. A question missed this way was undiagnosable after the fact. This is the smallest change that makes it visible and actionable.
Result	**Counter verified at 0 on clean streams; state transitions verified across a full speech cycle.**

4 · Regression Test Results

Suite	Scale	Baseline	Final	Verdict
VAD scored corpus	33 streams	95.7% recall / 50.0% FA	100% recall / 20.0% FA	**IMPROVED**
Gate + filter fingerprint	1,562 real transcripts	`7ac7d6aa…` kept 1517	`b15cdd0b…` kept 1518	**1 intended change**
End-to-end pipeline	6 scenarios	6/6, 1.30–1.33 s	6/6, 1.16–1.21 s	**FASTER**
Scenario behaviour suite	26 scenarios	28 answers, 1 cancel	28 answers, 1 cancel	**UNCHANGED**
Fault injection	5 modes + recovery	4/5 reported	5/5 reported	**IMPROVED**
Resource / lifetime	350 interactions	flat / 251 retained	flat / 251 retained	UNCHANGED
Answer quality (judged)	8 q × 7 rules	92.9% / 91.1%	96.4%	**≥ baseline**
Live full-stack server	real Groq STT + LLM	—	all stages reached	**PASS**
Multilingual	7 scripts	3 dropped	0 dropped	**FIXED**

4.1 Interruption / barge-in — unchanged, verified

```
barge_in_new_question                answers=2  cancels=1   cancel + new answer, no interleaving
fast_followup_interrupts_answer      answers=2  cancels=0   follow-up WAITS for history, no cancel
stray_word_during_answer             answers=1  cancels=0   small talk does not cancel
trailing_fragment_while_still_speaking answers=1  cancels=0   NEW: one question, one answer

Total real cancellation EVENTS across the whole 26-scenario suite: 1
```

A first pass appeared to show an extra cancellation in `fast_followup_interrupts_answer`. It was a false positive: the scenario's own note text contains the literal string "answer_CANCELLED". Re-counted against timestamped event lines only, giving the 1 above. Recording it because an unverified count would have been reported as a regression that did not exist.

4.2 Live full-stack pipeline (real uvicorn, real Groq)

```
audio → VAD → segment → STT → quality filter
  ready              sample_rate=16000 frame_ms=20
  speech_segment     2.86s (2.2s voiced) utt=1 final=True     VAD correct
  dropped            low_confidence :: 'Technical interview'   hallucination filter correct
  (synthetic formant tones are not speech; Whisper invented text from the
   vocabulary prompt and the filter caught it — the defence working end to end)

gate → LLM → answer socket  (via /ask)
  [ 0.05s] gate reason=manual_override ans=True
  [ 0.05s] answer_start model=openai/gpt-oss-120b
  [ 0.70s] FIRST TOKEN at 648ms
  [ 0.90s] answer_done 857ms, 476 chars                        real, correct answer
```

5 · Bugs Discovered During Implementation

Bug	Found by	Status
Negated character class excluded apostrophes, breaking English tokenisation for contractions	Gate fingerprint (`kept_by_filter` 1517→1516)	**FIXED before proceeding**
Comparison questions produce prose, not the bullets the prompt specifies	Live full-stack test	**Pre-existing, NOT a regression** — the 1,978-token prompt produced 0 bullets for the same question. Multi-part (3) and scenario (5) bullets identical before and after. Left alone.
The judged benchmark's `grounded` rule twice flagged résumé-present details as fabricated	Quality benchmark	Harness limitation, documented. Aggregate and token counts are sound; that one rule's numbers are noisy.
Scenario note in `stray_word_during_answer` claims "ANY new speech cancels" — the code no longer does that	Code reading	NOT FIXED — stale comment only, no behavioural impact. Listed for a future tidy-up.

6 · Not Implemented / Rejected / Future

Item	Status	Reason
STT provider racing	REJECTED	Per instruction, and per the audit: STT is 304 ms and not the bottleneck; racing doubles request rate into a rate-limited account.
WAV → FLAC for latency	REJECTED	Measured in the audit: STT latency is flat across a 5× payload range (47 KB → 250 KB, same latency). Network/fixed-overhead bound, not upload bound.
LLM connection warm-up	REJECTED	Measured in the audit: cold first call 452 ms vs warmed 472 ms. Hypothesis disproven.
Replacing Groq STT	REJECTED	Per instruction, and no evidence supports it.
Architecture rewrite	REJECTED	Per instruction. The dispatcher/consumer, gate architecture and barge-in were rated 8–9/10 and are untouched.
Streaming STT abstraction (§11)	FUTURE	**Inspected and deliberately not built.** The Riva adapter calls `offline_recognize` — it is batch, not streaming — and `riva.client` is not installed (commented out in `requirements.txt`). The blocker is not isolation: the `STTService` Protocol is `transcribe_pcm16(pcm) → TranscriptionResult`, which is inherently one-buffer-in / one-final-transcript-out and *cannot express* incremental partial results. Building an abstraction now would be speculative and would violate the "avoid unnecessary abstractions" instruction. **Concrete prerequisite for future work:** a second protocol method yielding partial hypotheses over an open connection, plus a gate that can consume partials.
Release per-session state (audit F-10)	NOT IMPLEMENTED	Measured impact is negligible: `session_id` defaults to `USERNAME`, a stable per-machine value, so entries are per-*user* (~324 bytes) not per-interview. 1,000 distinct users ≈ 324 KB. The change also carries real risk (orphaning an in-flight answer when the audio socket closes before the answer socket). Deferred until session ids become per-interview.
Remove dead LangGraph path (audit F-11)	NOT IMPLEMENTED	Out of the approved scope for this pass. Provably unreachable, zero runtime effect. Safe to remove whenever wanted.
Lower `SEGMENT_END_SILENCE_MS`	NOT IMPLEMENTED	The audit advises against it as a first move, and that advice still holds: 420 ms already splits utterances at exactly that boundary, and shortening it produces *more* splitting. Now that F-03 is in, it could be revisited with measurement — but only after real-audio validation.
Multilingual rule gate	FUTURE	F-04 fixes the *filter*, so Hindi now reaches the gate. But `_QUESTION_PREFIXES`, `_SMALL_TALK` and `_INTERVIEW_INTENTS` remain English-only, so non-English questions route to the tier-2 LLM classifier (+392 ms) instead of matching a free rule. Correct but slower. Needs a product decision on which languages are in scope.

7 · Success Criteria

Criterion	Result	Evidence
Speech detection improves or stays stable	**IMPROVED**	95.7% → 100% recall
Noise false triggers decrease	**IMPROVED**	50% → 20%
Question splitting decreases	**FIXED**	log-replayed case: 2 answers → 1
LLM token consumption decreases	**IMPROVED**	2,719 → 1,970 per question
LLM throttling decreases	**IMPROVED**	2.9 → 4.1 questions/minute
First response latency improves or stays stable	**IMPROVED**	1.32 s → 1.18 s
STT accuracy does not regress	**HELD**	1,517 English transcripts identical
English behaviour does not regress	**HELD**	1,561/1,562 fingerprint identical
Multilingual behaviour improves	**IMPROVED**	Devanagari/Tamil/CJK now reach the gate
Interruption handling remains functional	**HELD**	28 answers, 1 cancel — identical
Error recovery remains functional	**IMPROVED**	5/5 modes reported, was 4/5
No new race conditions	**HELD**	Suite unchanged; `_is_settled` reads under the existing lock
No duplicate answers	**IMPROVED**	The one known duplicate source is fixed
No stuck states	**HELD**	Fault injection: no stuck task in any mode

8 · Production Readiness

Area	Audit	Now	Basis
Speech detection	4	8	100% recall, 20% FA, plus telemetry where there was none
STT accuracy	6	7	Multilingual unblocked; still no WER on real speech
Question understanding	8	9	Splitting fixed; classification provably unchanged
LLM answer quality	8	8	96.4% judged, at 28% fewer tokens
First-response latency	6	8	−140 ms, and the stall cliff moved from Q3 to Q4–5
Interruption handling	9	9	Preserved exactly
Error recovery	8	9	Last silent path closed
Reliability	6	7	Throttling reduced but not eliminated — 4.1 q/min is still a real ceiling
Observability	6	8	The invisible failure now logs a WARNING naming the trigger level
Resource efficiency	8	8	Unchanged; log I/O 19× cheaper
Code quality / maintainability	8	8	Every new constant carries its measurement, per project convention
OVERALL	6.9	**8.2**	**All three headline defects closed**

The one thing still capping this system

4.1 questions per minute is better than 2.9, but it is still a ceiling a busy interview can hit. The prompt work took what could be taken safely; the remaining lever is commercial, not technical. A higher Groq tier removes the constraint entirely with no code change. Alternatively, `AnthropicLLMClient.stream_chat` already implements prompt caching correctly — the system block is sent with `cache_control: ephemeral` as a proper cacheable prefix, and an `ANTHROPIC_API_KEY` is already configured — which would take the 1,186-token prompt largely out of the per-minute budget after the first call. That path is **unvalidated**: quality and TTFT on `claude-haiku-4-5` were not benchmarked here, and should be before switching.

9 · Recommended Next Steps

1. **Validate the VAD threshold against real recorded call audio.** The 33-stream corpus is synthetic two-formant tones — correct for characterising an energy detector, but the absolute RMS levels should be confirmed against your own loopback audio at several system volumes before you rely on 150 in a real interview. This is the highest-value outstanding validation.
2. **Watch for the new sub-threshold WARNING.** If it appears in a real session, the threshold is still too high for that line and you now have the number to tune against.
3. **Resolve the token ceiling commercially** — higher Groq tier, or benchmark the Anthropic path whose caching is already wired.
4. **Build a WER dataset** (30–50 recorded questions with reference transcripts, including Indian-accented English, technical terms and names). STT accuracy remains the weakest-evidenced area of both reports.
5. **Decide the multilingual scope.** The filter is fixed; the rule gate is not. If Hindi matters, add Hindi question prefixes so it stops paying the tier-2 round trip.

Method statement. Baseline was captured from the untouched `7a375cf` tree and verified against the audit's recorded fingerprints before the first edit. The gate corpus was frozen to a file so that log growth during implementation could not move the baseline. Every change was regression-tested before the next was started; the one regression introduced (apostrophe tokenisation) was caught by the fingerprint gate and fixed before proceeding, per the "do not stack fixes on a broken change" rule. All figures in this report are measured; no result is estimated, extrapolated or carried over from the audit without re-measurement. Where a number could not be measured it is named as such.